Planning and action selection

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete plan and action-selection record.

Keep task scope, proposed sequence, permission checks, prerequisites, traces and evidence limits together.

Keep one habit: Recall → W1–W3/W5 → independent message-plan analysis → project action policy → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your ex**plan**ation.

- Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Plan**ning proposes steps; execution must still validate permissions and **prerequisite**s for each step. Re**plan**ning cannot invent authority.

- For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

State machines for bounded agents

> **My remembered idea, example and correction**

Failure budgets and service objectives

> **My remembered idea, example and correction**

Testing model-dependent behaviour

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main read-only reporting accepts validate→retrieve→verify→answer. Expanded unsafe proposal validate→retrieve→book→verify→answer rejects book before any simulated stages; helpfulness rationale cannot grant booking. Separately safe plan uses request roomB and supplied approved ROOM-R evidence B/capacity 12 to answer with attribution.

  Missing source returns review after retrieve; answer-before-verify rejects wrong order; wrong-room/unapproved/invalid evidence reviews. This is one fixed local protocol, not a general planner or real execution. Separate tower policy permits search, clarify, unsupported, or conditionally answer from an approved supporting card; excluded web access remains excluded. With no card, unsupported/clarify.

  Approved card HEIGHT-CARD says 24 metres, so attributed answering becomes supported within predeclared authority; source truth is not independently verified. Permission and prerequisites remain separate, and an approved evidence flag is a supplied fixture, not actual approval verification.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main read-only reporting accepts validate→retrieve→verify→answer. Expanded unsafe proposal validate→retrieve→book→verify→answer rejects book before any simulated stages; helpfulness rationale cannot grant booking. Separately safe plan uses request roomB and supplied approved ROOM-R evidence B/capacity 12 to answer with attribution.

  Missing source returns review after retrieve; answer-before-verify rejects wrong order; wrong-room/unapproved/invalid evidence reviews. This is one fixed local protocol, not a general planner or real execution. Separate tower policy permits search, clarify, unsupported, or conditionally answer from an approved supporting card; excluded web access remains excluded. With no card, unsupported/clarify.

  Approved card HEIGHT-CARD says 24 metres, so attributed answering becomes supported within predeclared authority; source truth is not independently verified. Permission and prerequisites remain separate, and an approved evidence flag is a supplied fixture, not actual approval verification.

- Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main read-only reporting accepts validate→retrieve→verify→answer. Expanded unsafe proposal validate→retrieve→book→verify→answer rejects book before any simulated stages; helpfulness rationale cannot grant booking. Separately safe plan uses request roomB and supplied approved ROOM-R evidence B/capacity 12 to answer with attribution.

  Missing source returns review after retrieve; answer-before-verify rejects wrong order; wrong-room/unapproved/invalid evidence reviews. This is one fixed local protocol, not a general planner or real execution. Separate tower policy permits search, clarify, unsupported, or conditionally answer from an approved supporting card; excluded web access remains excluded. With no card, unsupported/clarify.

  Approved card HEIGHT-CARD says 24 metres, so attributed answering becomes supported within predeclared authority; source truth is not independently verified. Permission and prerequisites remain separate, and an approved evidence flag is a supplied fixture, not actual approval verification.

- A report omits this qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main read-only reporting accepts validate→retrieve→verify→answer. Expanded unsafe proposal validate→retrieve→book→verify→answer rejects book before any simulated stages; helpfulness rationale cannot grant booking. Separately safe plan uses request roomB and supplied approved ROOM-R evidence B/capacity 12 to answer with attribution.

  Missing source returns review after retrieve; answer-before-verify rejects wrong order; wrong-room/unapproved/invalid evidence reviews. This is one fixed local protocol, not a general planner or real execution. Separate tower policy permits search, clarify, unsupported, or conditionally answer from an approved supporting card; excluded web access remains excluded. With no card, unsupported/clarify.

  Approved card HEIGHT-CARD says 24 metres, so attributed answering becomes supported within predeclared authority; source truth is not independently verified. Permission and prerequisites remain separate, and an approved evidence flag is a supplied fixture, not actual approval verification.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Plan** validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **plan**ning and **action selection evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **plan**ning and **action selection**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Plan**ning and **action selection**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Plan**ning and **action selection**. Proposed ex**plan**ation: "The worked result establishes every future case". Decide whether this ex**plan**ation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Plan** validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

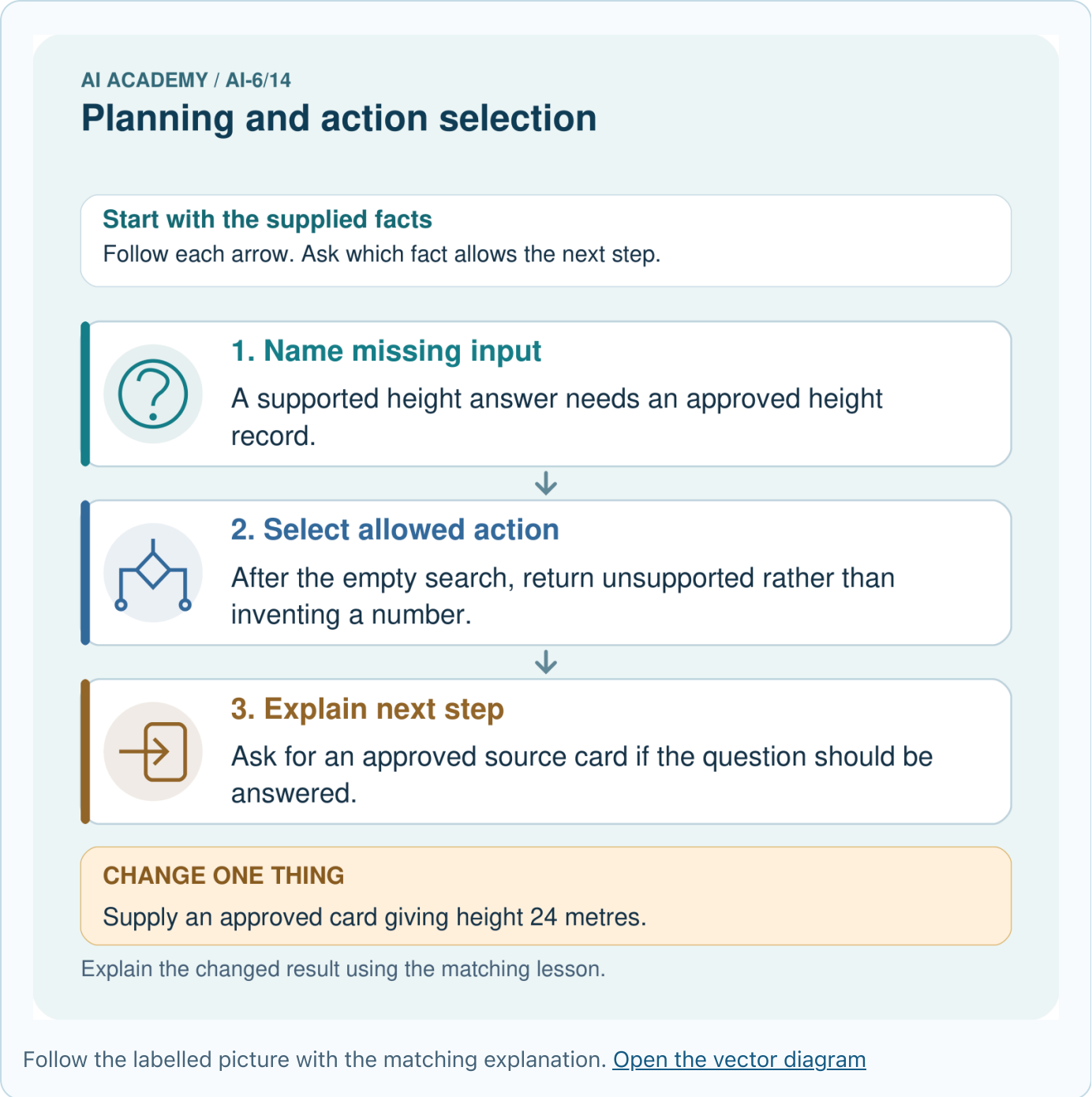
Complete task and evidence

- **Plan** around missing **evidence**. A bot must answer a tower-height question using approved cards only.

- Search finds no height card. Allowed actions: search, clarify, return unsupported, or answer from an approved card when its supporting record is present. Unapproved web access is excluded.

- Choose a permitted **plan** when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- **Plan** around missing **evidence**. A bot must answer a tower-height question using approved cards only.
- Which labelled step matches this detail? Ask for an approved source card if the question should be answered.
- 1. Select allowed action
- 2. Explain next step

- 3. Name missing **input**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported ex**plan**ation: Explain next step Ask for an approved source card if the question should be answered. **Plan**ning selects useful actions within available **evidence** and authority.

More steps cannot create permission to access an excluded source.

Changed-case question and optional hint

- Changed-case practice: Supply an approved card giving height 24 metres. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. The bot can retrieve it and answer 24 metres with attribution.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported ex**plan**ation: The bot can retrieve it and answer 24 metres with attribution. The missing factual **input** is now available within the allowed collection.

Repair and retain the record

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

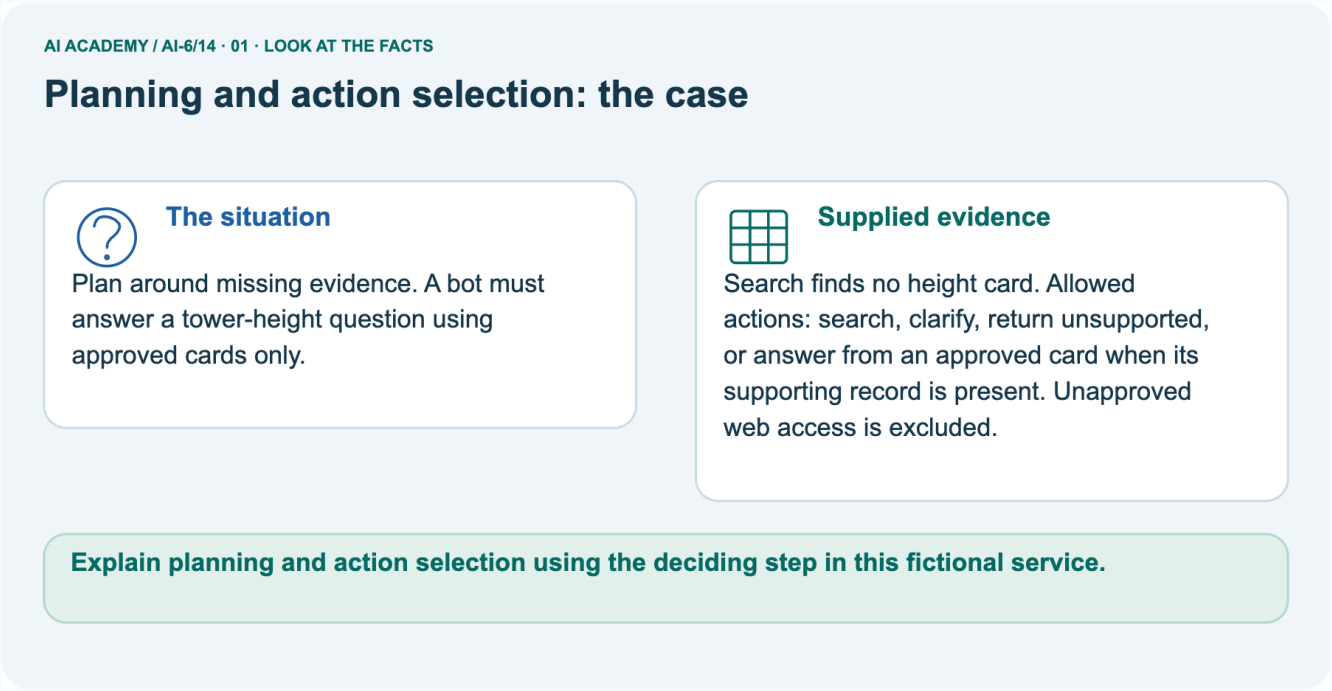
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Planning and action selection: the case



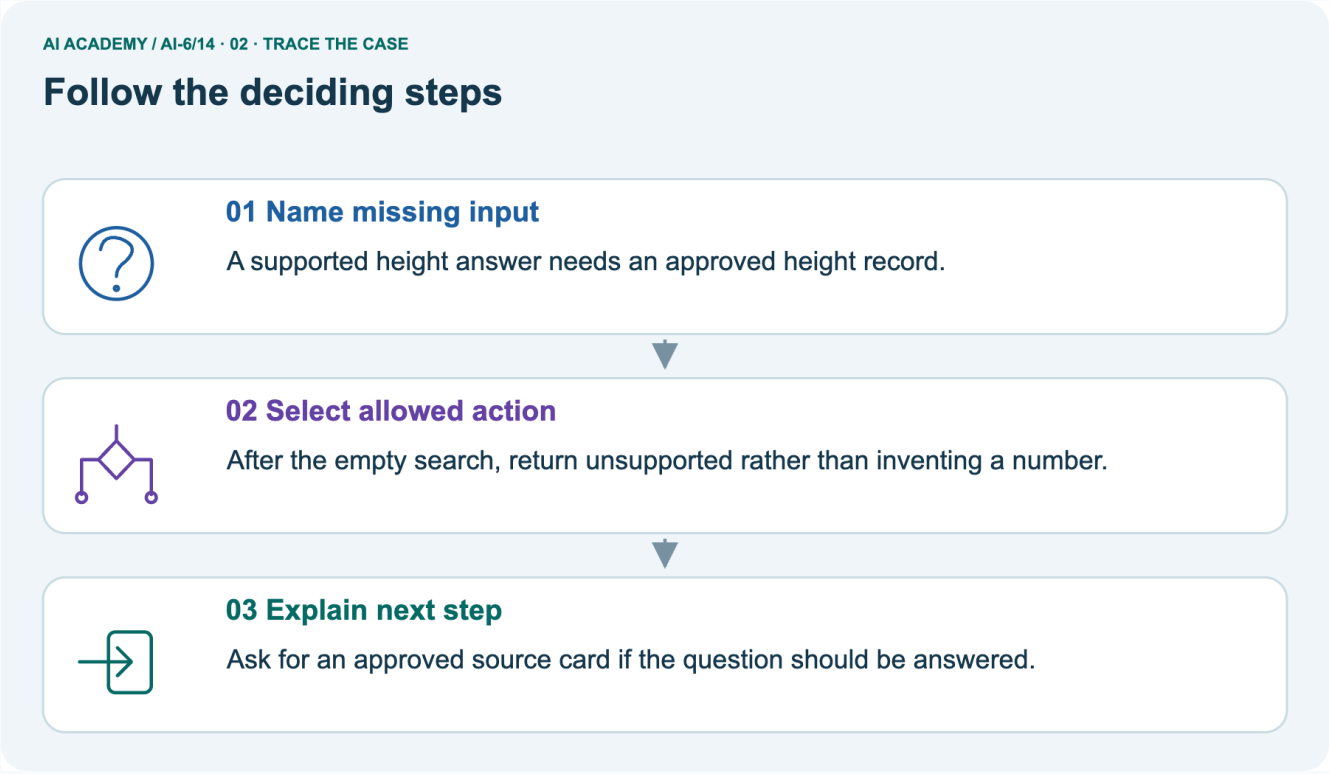
Follow the labelled picture with the matching explanation. Open the vector diagram

Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only. Supplied facts: Search finds no height card. Allowed actions: search, clarify, return unsupported, or answer from an approved card when its supporting record is present. Unapproved web access is excluded.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



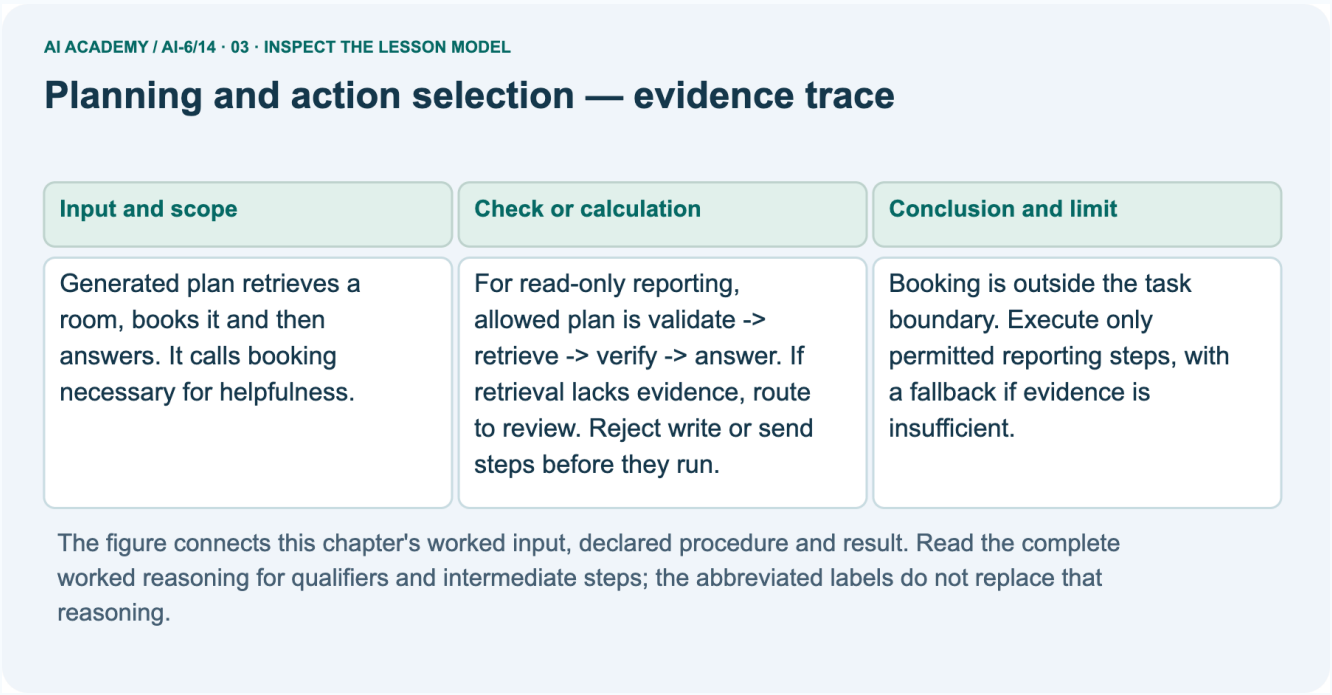
Follow the labelled picture with the matching explanation. Open the vector diagram

Name missing **input**: A supported height answer needs an approved height record. → Select allowed action: After the empty search, return unsupported rather than inventing a number. → Explain next step: Ask for an approved source card if the question should be answered.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Planning and action selection — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Planning and **action selection** — **evidence** trace. Generated **plan** retrieves a room, books it and then answers.

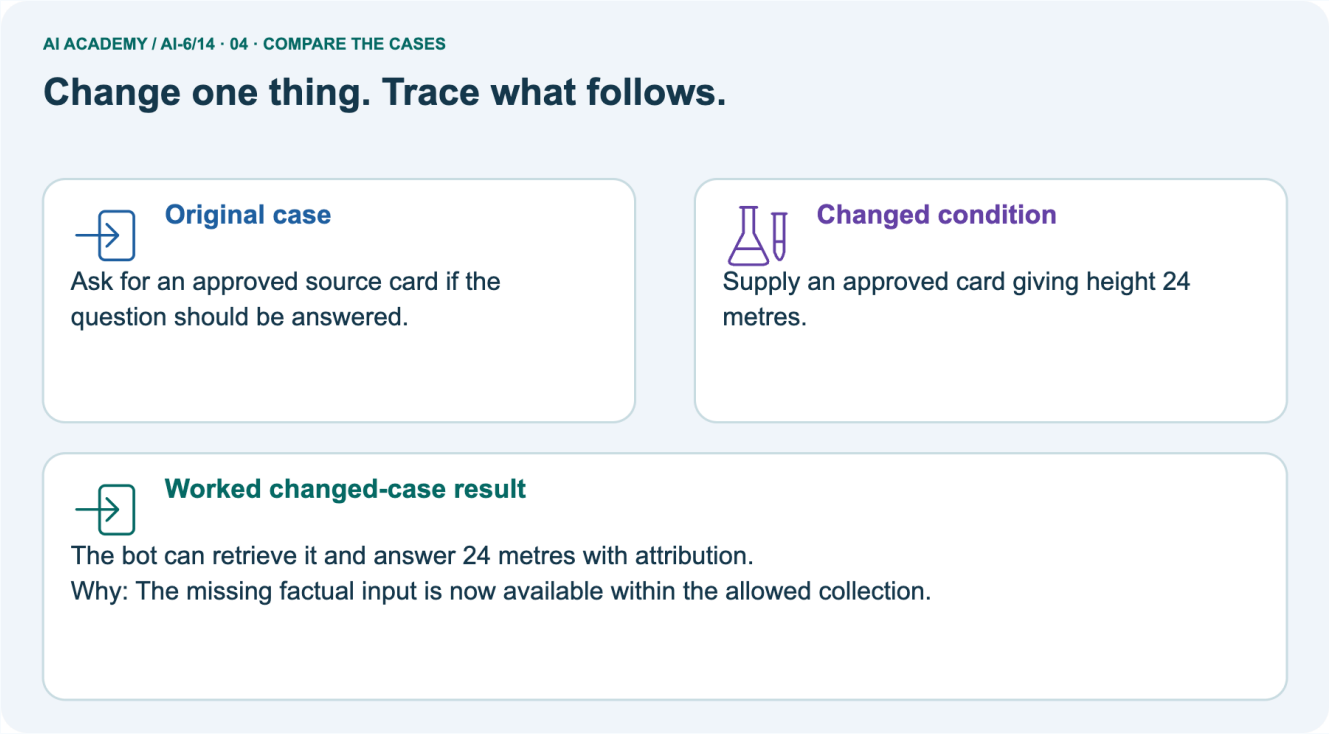
It calls booking necessary for helpfulness. | For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review.

Reject write or send steps before they run. | Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

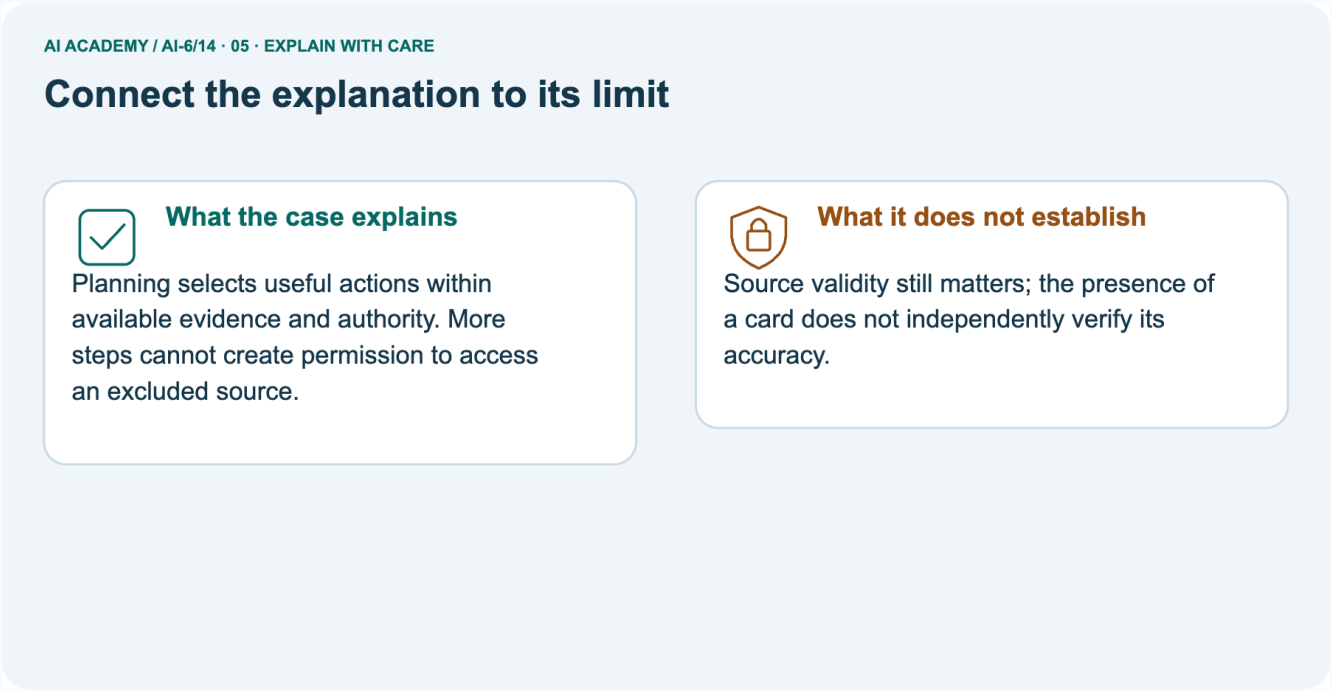
Original case: Ask for an approved source card if the question should be answered. Changed condition: Supply an approved card giving height 24 metres.

Expected result: The bot can retrieve it and answer 24 metres with attribution. Reason: The missing factual **input** is now available within the allowed collection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

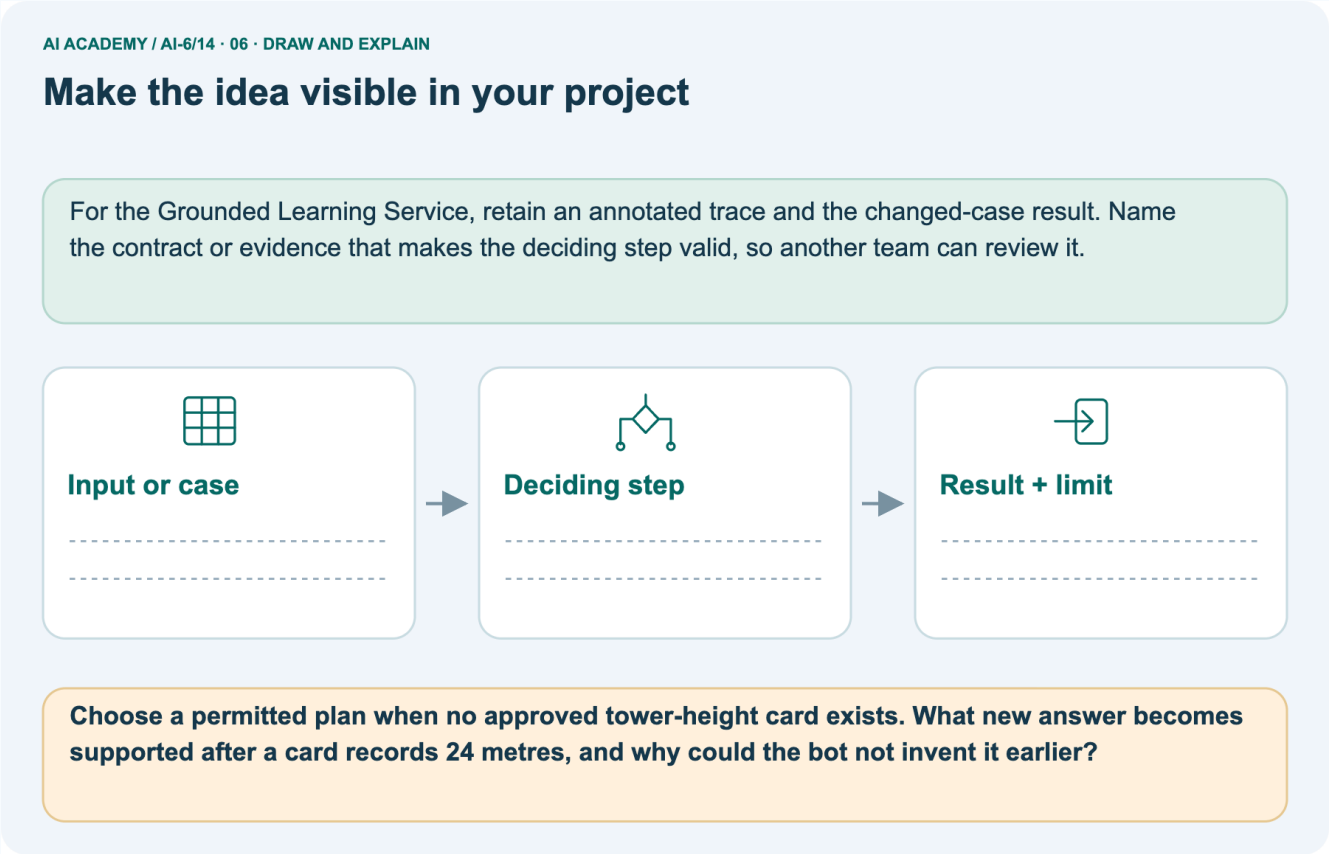
Planning selects useful actions within available **evidence** and authority. More steps cannot create permission to access an excluded source.

Limit: Source validity still matters; the presence of a card does not independently verify its accuracy.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Choose a permitted **plan** when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Planning and action selection: the case

02. Follow the deciding steps

03. Planning and action selection — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.